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Sensitivity Analysis of a TIR-Based Cold-Water Patch Detection Algorithm

The Influence of Cold Water Tributaries on Cold Water Patches

Bachelor Thesis

by
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Bachelor program Applied digital Life Sciences
Focus Digital Environment
Study cohort 2023

June 22, 2026

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Imprint

Keywords:

Thermal Infrared (TIR), Cold-Water Patches (CWP), Sensitivity Analysis, Morris Method, River Temperature, Remote Sensing, GDAL, R

Suggested Citation:

Dürig, E. (2026). Sensitivity Analysis of a TIR-Based Cold-Water Patch Detection Algorithm. Bachelor Thesis, ZHAW School of Life Sciences and Facility Management.

Institute:

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Chapter 1

Introduction

In recent years, thermal infrared (TIR) remote sensing from unmanned aerial vehicles (UAVs) and other airborne platforms has become an established approach for mapping thermal heterogeneity in river systems (Casas-Mulet et al., 2020; Dugdale, 2016; Kuhn et al., 2021; Torgersen et al., 2001). Mapping this heterogeneity is ecologically important: the cold-water patches it reveals can act as refugia that are essential for the survival of thermally sensitive fish species, particularly as river temperatures in Switzerland are expected to rise due to climate change. Despite this importance, there is currently no comprehensive information about the presence, scale, and location of such cold-water patches in Swiss water bodies (Tonolla and Antonetti, 2021, 2026).

As part of a cantonal project aiming to quantify the temperature heterogeneity and detect cold-water patches in the Emme River and its tributaries, a helicopter-based survey was previously conducted to collect TIR imagery covering over a 100 km river transect. These TIR images were processed using a newly developed algorithm for the automatic detection of cold-water patches, extracting longitudinal and transversal temperature gradients alongside the detected cold-water spots. Detailed information regarding the full workflow can be found in (Tonolla and Antonetti, 2021, 2026); the present thesis builds directly on this existing algorithm and the resulting dataset.

The CWP detection algorithm relies on externally configured parameters, and sensitivity analyses performed in (Tonolla and Antonetti, 2026) indicate that the detection results can vary significantly based on the chosen parameter values, with the step length and the temperature delta parameters having a particularly strong influence. The algorithm divides the river reach into fixed-length segments and computes a separate local reference temperature for each segment; visual inspection of existing detection results revealed that this can produce abrupt cut-off artefacts, where a cold-water patch spanning two adjacent segments is compared against two different reference temperatures and truncated at the segment boundary, even though the underlying thermal anomaly continues beyond it. Because cold-water tributaries tend to produce elongated thermal anomalies extending substantially along the river channel, such patches are disproportionately likely to span a segment boundary, suggesting that step length sensitivity may be elevated specifically for tributary-caused CWPs. A systematic and more extensive sensitivity analysis of the algorithm is therefore needed to test this assumption, to better characterise the resulting variance more broadly, and to inform potential algorithmic improvements. Performing such an analysis requires executing the algorithm across a large number of parameter combinations. The existing algorithm was found to require between 1.25 and 2 hours per

run on the available hardware, rendering a full sensitivity analysis infeasible without a faster implementation. A speed-optimized reimplementation is therefore a prerequisite for the sensitivity analysis itself.

Research Questions

Based on the above, this thesis pursues three objectives: first, to investigate whether the existing CWP detection algorithm can be reimplemented in a computationally more efficient form while producing equivalent outputs; second, to systematically investigate the algorithm's sensitivity to its parameters, with particular focus on how the presence of tributary inflows interacts with this sensitivity; and third, to derive adjustments to the detection algorithm that could reduce this parameter sensitivity while maintaining ecologically plausible detection results. These objectives are formalised in the following three research questions:

Research question 1 Can the existing CWP detection algorithm be reimplemented in a computationally more efficient form, while producing outputs that are equivalent to those of the original implementation?

Research question 2 How does the step length parameter contribute to the variance of the detected CWP area for tributary-caused CWPs, compared to non-tributary-caused CWPs?

H_0 The sensitivity of the detected CWP area to the step length parameter does not differ between tributary-caused and non-tributary-caused CWPs.

H_1 The sensitivity of the detected CWP area to the step length parameter is significantly higher for tributary-caused CWPs than for non-tributary-caused CWPs.

Research question 3 Building on the findings of [Tonolla and Antonetti \(2026\)](#) sensitivity analysis: what adjustments could be made to the detection algorithm to reduce variability with respect to its parameters while also achieving plausible detection results?

Chapter 2

Theoretical Background

2.1 Sensitivity Analysis

According to Saltelli et al. sensitivity analysis is “The study of how uncertainty in the output of a model [...] can be apportioned to different sources of uncertainty in the model input” (Saltelli, 2004). This definition rests on a broad notion of what counts as a model input: in sensitivity analysis, everything that causes variation in the output of the model is classified as an input (Saltelli et al., 2007). Providing an example: if a linear regression of a target variable Y is performed, then not only are the predictors $X_1, \dots, X_n$ considered as model inputs, but also the parameters $\theta_1, \dots, \theta_n$ that correspond to the predictors. So the model:

$$Y = f(X_1, X_2, \theta_1, \theta_2) \tag{1-1}$$

would have four model inputs from the view of sensitivity analysis.

Sensitivity analysis provides a range of benefits. Firstly it allows the modeller to apply a strategy called “Factor Fixing”: if the model output Y is found to be independent of one or multiple inputs X_i , then these can be set to fixed constants, making sensitivity analysis a tool for model simplification (Saltelli et al., 2007). Secondly, sensitivity analysis generally improves model validity by identifying highly influential and therefore critical model components (Ligmann-Zielinska, 2013).

2.2 The Morris Method

The Morris method belongs to the group of screening methods in sensitivity analysis. Screening methods require only a small number of model evaluations, making them computationally cheap and therefore well suited to models that are expensive to run (Campolongo et al., 2007; Saltelli et al., 2007). The resulting sensitivity indices cannot be interpreted quantitatively, but they do allow for a qualitative, ordinal ranking of which model inputs are more important than others (Campolongo et al., 2011). As a global method, Morris explores the entire input space by varying all input parameters simultaneously across their plausible ranges, meaning that sensitivity scores reflect average behaviour across the full parameter space rather than being tied to a single point (Saltelli and Annoni, 2010). The following explanation of the Morris method closely follows (Saltelli et al., 2007) and is

explained more explicitly in chapter 3 of their book.

2.2.1 Elementary Effects and the Sampling Grid

At the core of the Morris method sits a k -dimensional model Y with inputs $X_1, \dots, X_k$. The model inputs are defined in the k -dimensional unit cube, meaning that X_i can take on a value in the interval $[0, 1]$. It is worth pointing out, that every model with a continuous input can be scaled to the $[0, 1]$ range, which makes the unit cube a great place to define methods. In the Morris method, this input space is discretized into a grid Ω , which is used as a basis for the computation of so-called elementary effects.

An elementary effect of the i th input parameter X_i is defined as:

$$EE_i = \frac{Y(X_1, X_2, \dots, X_{i-1}, X_i + \Delta, \dots, X_k) - Y(X_1, X_2, \dots, X_k)}{\Delta} \quad (2-2)$$

Where Δ represents some step along the grid Ω . Figure 2-1, taken from (Saltelli et al., 2007), shows an example of such a grid for a two-parameter, four-level case.

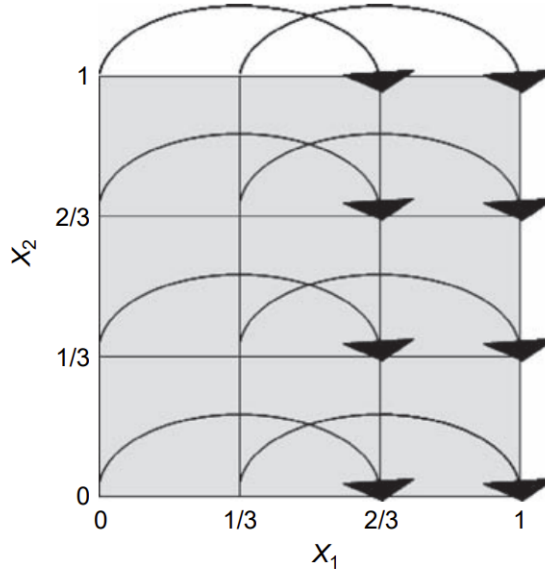


Fig. 2-1: An example grid Ω . The grid features two parameters X_1, X_2 , meaning that it is a two-dimensional grid. Further, it features four levels ($p = 4$). The step length Δ chosen here is $\frac{2}{3}$ and indicated by the arrows.

The Morris method occupies itself with finding the statistical distribution of each of the i elementary effects. $\mathcal{F}_i$ is thereby used to denote the distribution of EE_i , in other words $\mathcal{F}_i \sim EE_i$ (Saltelli et al., 2007).

2.2.2 Trajectory-Based Sampling

To populate $\mathcal{F}_i$ with multiple samples of EE_i , the Morris method uses an efficient trajectory-based sampling design that obtains as many elementary effects EE_i from as few model evaluations as possible. The exact mathematical formulation of how trajectories are constructed is beyond the scope of this thesis; a visual understanding is sufficient. Figure 2-2,

taken from (Saltelli et al., 2007), shows an example trajectory through a three-dimensional sampling grid.

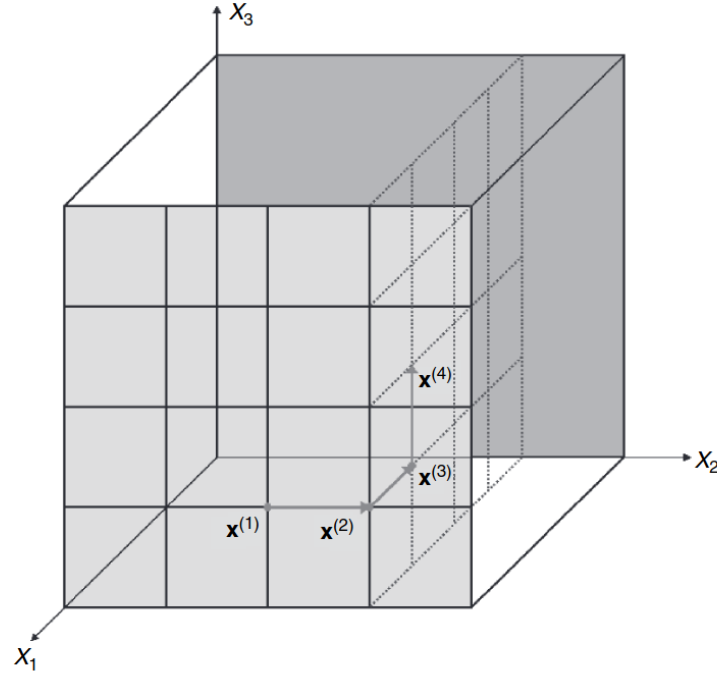


Fig. 2-2: An example Morris trajectory through a three-dimensional sampling grid ($k = 3$, $p = 6$). The trajectory consists of $k + 1 = 4$ points.

From this figure it becomes visible that a trajectory consists of $k+1$ points $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(k+1)}$, where k is the dimension of the sampling grid. Each point differs from its predecessor by a step of Δ in exactly one grid direction, so that consecutive points differ in only one model input X_i . The trajectory is constructed such that each dimension of the grid is stepped along exactly once. Each trajectory yields exactly k elementary effects EE_i – one for each parameter spanning the sampling grid – and requires $k + 1$ model evaluations.

To build up $\mathcal{F}_i$ for every factor, r such trajectories are generated. Since each trajectory contributes exactly one elementary effect per factor, each $\mathcal{F}_i$ ends up containing r samples. The total number of model evaluations required therefore is:

$$N = r \cdot (k + 1) \quad (2-3)$$

(Saltelli et al., 2007).

2.2.3 Choosing Δ and r

The choice of r and Δ is not independent of p . When p is chosen to be even, the choice $\Delta = \frac{p/2}{p-1}$ is ideal, as it has the additional property of guaranteeing equal-probability sampling from the grid, meaning that all grid levels are sampled with equal probability across the r trajectories (Saltelli et al., 2007). There is no clear rule regarding the number of trajectories r to pick. Generally, as the number of levels p in the grid increases, the number of trajectories r also needs to increase. Otherwise, increasing p serves little purpose, as the grid will not be sufficiently explored to feature all levels if the number of trajectories

is too small (Saltelli et al., 2007). Generally, the number of trajectories is usually chosen somewhere between 10–50 (Campolongo et al., 2007).

2.2.4 Sensitivity Measures: μ , σ , and μ^*

Once the distributions $\mathcal{F}_i$ have been mapped out, statistical measures can be computed, specifically the mean μ_i and the standard deviation σ_i of $\mathcal{F}_i$ are of interest. These two statistical measures together already allow for reasonable insight into the influence of the i th parameter X_i , which corresponds to the i th elementary effect EE_i , onto the model:

If μ_i deviates strongly from 0, then this means that the elementary effect EE_i generally takes on non-zero values and therefore that the model output changes notably when a step of Δ is performed along the dimension spanned by X_i . This ultimately means that changes in X_i generally affect the model output Y considerably, and that X_i therefore receives a high sensitivity score.

If σ_i is large, this means that the distribution $\mathcal{F}_i$ is spread out considerably over a larger range. This indicates that the computed elementary effects EE_i generally differed depending on their location in the grid. This behaviour points either to nonlinearity regarding X_i or to the fact that X_i interacts with other input parameters. Which of the two effects is responsible for the high standard deviation – or whether both contribute simultaneously – cannot be determined, which is a shortcoming of the Morris method (Saltelli et al., 2007). Figure 2-3 provides a visual example of possible $\mathcal{F}$ along with the associated μ and σ values and their corresponding interpretation.

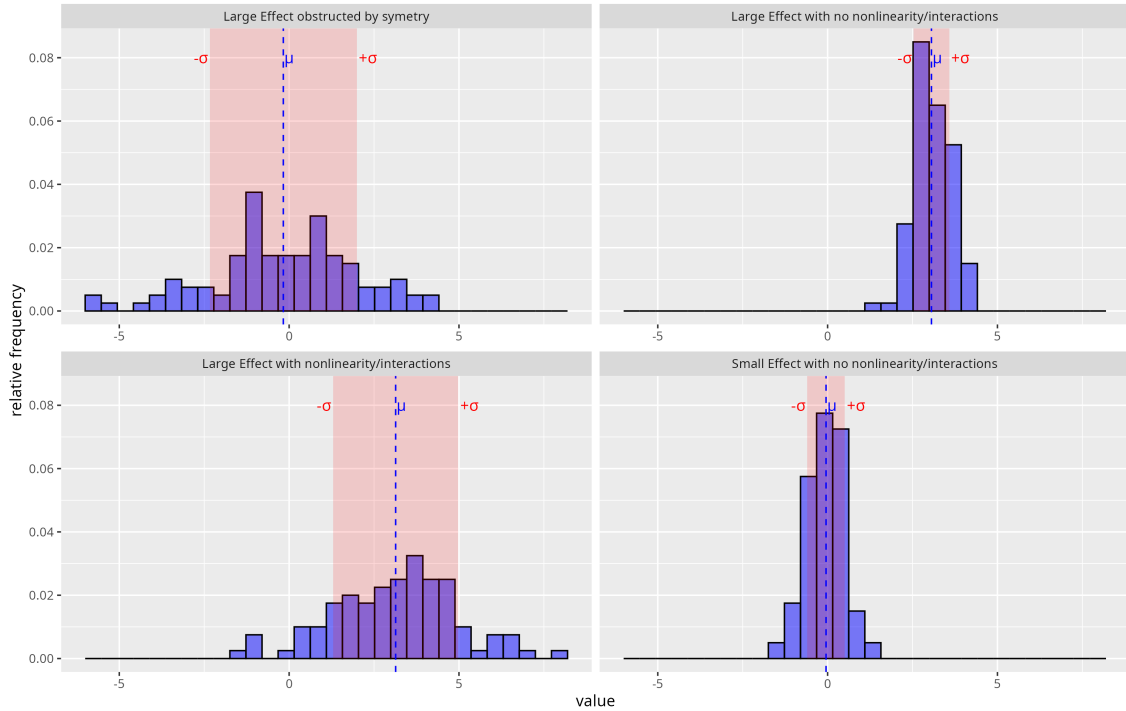


Fig. 2-3: Four example distributions $\mathcal{F}_i$ of a parameter X_i , illustrating how μ and σ jointly characterise parameter influence. *Top left:* a large effect obscured by symmetry. *Top right:* a large effect with no nonlinearity or interactions. *Bottom left:* a large effect with nonlinearity or interactions. *Bottom right:* a small effect with no nonlinearity or interactions.

Situations can occur where $\mathcal{F}_i$ contains many extreme values, yet the distribution is symmetric around zero. One such example is visualized in the upper-left plot of Figure 2-3. In such cases, computing the mean value μ_i yields a value close to zero, leading to the assumption that the parameter X_i is noninfluential. This would be a faulty assumption, as the sampled elementary effects EE_i are substantial in magnitude but cancel each other out. This phenomenon can be detected by always taking σ_i into consideration, since in this scenario σ_i is large. It is, however, more common to additionally sample a distribution $\mathcal{G}_i$, which captures the magnitude of the elementary effects $|EE_i|$. We then compute the mean of this distribution, μ_i^* , which can be interpreted like μ_i . Since it only captures magnitude, this distribution however never features negative values, and cancellation effects can therefore not occur (Saltelli et al., 2007). Figure 2-4 provides a visual example of this mechanism.

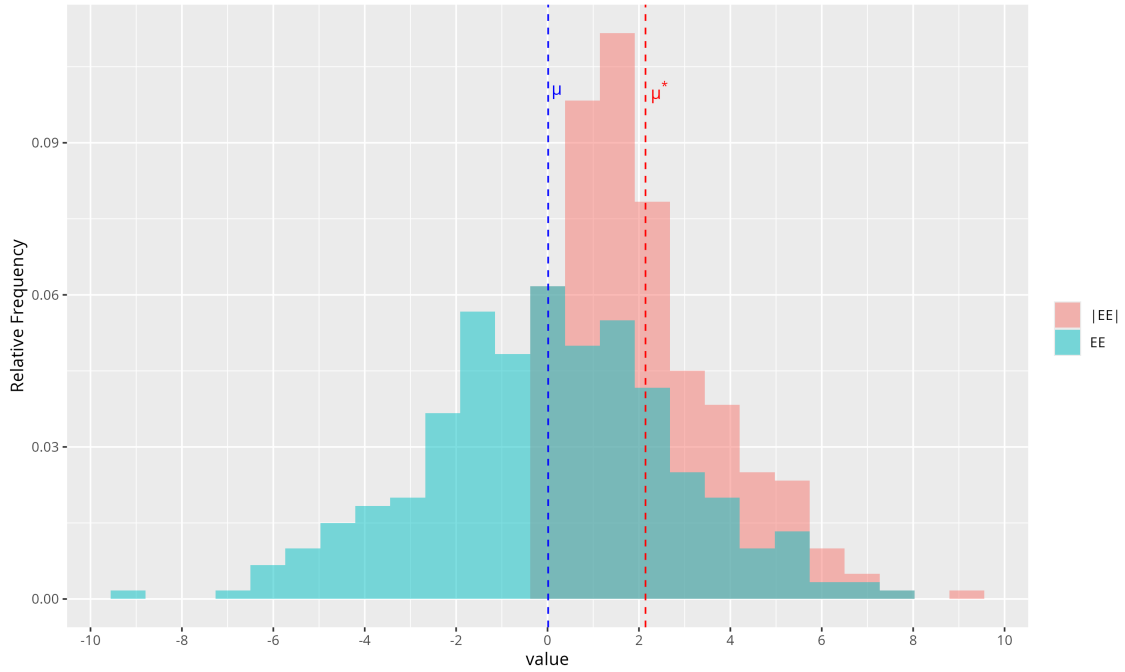


Fig. 2-4: The difference of μ and μ^* visualized with their underlying distribution $\mathcal{F}_i$ and $\mathcal{G}_i$. While the μ lies at around zero, μ^* significantly deviates from zero.

2.3 Application to Spatial Models

The Morris method, like most global sensitivity analysis methods, is defined for models with a scalar output: a single value Y is required for each model evaluation (Saltelli et al., 2007). The CWP detection algorithm, however, produces a spatially distributed output. Applying the Morris method to such a model therefore requires a spatial aggregation step to aggregate the spatial output to a scalar value, e.g. using a statistical aggregation function such as the mean (Lilburne and Tarantola, 2009).

2.4 Coldwater Patches

Cold water patches (CWPs) are areas in rivers and streams where the water temperature is measurably colder than that of the main channel. The temperature difference used to

define a CWP varies across the literature but is typically set between 0.5 and 3 degrees Celsius below the main channel temperature ([Casas-Mulet et al., 2020](#); [Wawrzyniak et al., 2016](#)). Cold water patches are not automatically cold water refugia. Cold water refugia are cold water patches that are actively used by cold-water organisms for thermal relief ([Mejia et al., 2023](#)). The ecological importance of cold water refugia is expected to increase as river temperatures rise due to climate change ([Kurylyk et al., 2015](#)). CWPs can arise from a range of physical mechanisms. A common cause is the inflow of a colder tributary-. Other generating mechanisms include groundwater upwelling, lateral seeps or hyporheic exchange ([Mejia et al., 2023](#)).

Chapter 3

Methods

This chapter describes the operationalisation of the three research questions into a set of methodological steps, which are then described in detail. The first section addresses research question 1 by describing the reimplementation of the CWP detection algorithm and the comparison of its outputs and runtime against the original. The second section addresses research question 2 by describing the Morris sensitivity analysis, including the selection and classification of CWPs, the computation of sensitivity indices, and the subsequent visualisation and statistical modelling. The third section addresses research question 3 by describing the algorithm modifications derived from the sensitivity analysis findings. A description of the data and study area is provided first.

3.1 Data and Study Area

The Emme river originates in an extensive moorland between the Hohgant and Augstmatthorn massifs and flows through the Emmental region before joining the Aare near Zuchwil/Luterbach. The catchment covers approximately 974 km² with a mean elevation of 845 m a.s.l., and land cover is dominated by agricultural land (52.1%) and forested areas (37.9%). During summer thunderstorms, the river can rise rapidly within minutes and transport large quantities of driftwood; as a consequence, the lower reaches have been heavily engineered ([Tonolla and Antonetti, 2026](#)).

[Tonolla and Antonetti \(2026\)](#) acquired high-resolution TIR and RGB orthophotos along a 69 km transect of the Emme main channel, spanning from Kemmeribodenbad BE to Biberist SO, as well as imagery of three tributary rivers: the Ilfis, Grüene, and Urtene. The TIR imagery was collected from a helicopter platform using a FLIR SC655 thermal infrared camera. In this thesis, only the TIR and RGB orthophotos of the 69 km main channel section are used. The TIR orthophotos serve as the primary input to the CWP detection algorithms, while the RGB orthophotos are used in a downsampled state as background images for selected visualisations and were not used for any further analytical purpose. The 69 km river transect is covered by three partially overlapping single-band TIR orthophotos. [Figure 1-1](#) provides an overview of the study area while [Table 1-1](#) summarises their technical specifications of the TIR orthophotos.

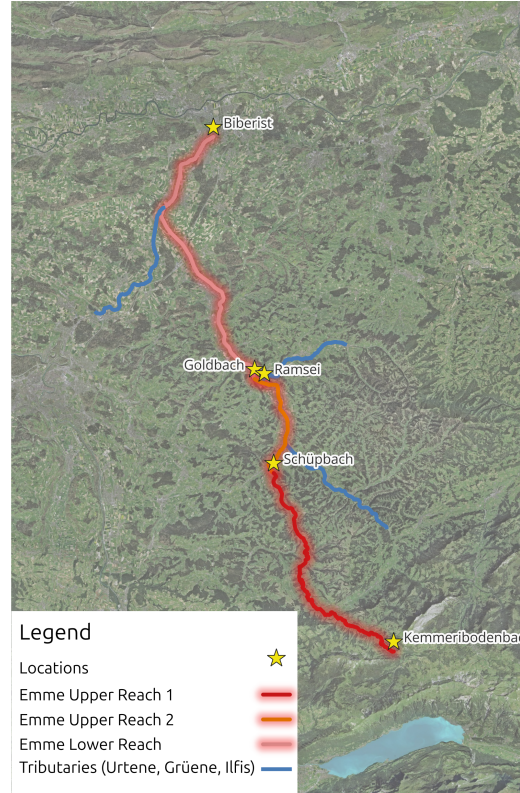


Fig. 1-1: Overview of the study area along the Emme river. The map shows the settlements bounding each of the three TIR orthophoto segments (Emme Upper Reach 1, Emme Upper Reach 2, Emme Lower Reach), as well as the three tributary rivers imaged during the survey (Ilfis, Grüene, and Urtene), shown here as a single combined category.

Tab. 1-1: Technical specifications of the three TIR orthophotos covering the Emme main channel. All rasters are in the Swiss coordinate reference system CH1903+/LV95 (EPSG:2056) and stored as single-band Float32 GeoTIFF files.

	Emme Reach 1	Upper Emme Reach 2	Upper Emme Reach	Lower Emme Reach
Extent	Kemmeribodenbad – Schüpbach	Schüpbach – Goldbach	Ramsei – Biberist	
Total pixels	4.75×10^9	8.45×10^8	6.57×10^9	
Resolution (m)	0.207×0.207	0.204×0.204	0.207×0.207	
Min temp. (°C)	7.43	10.08	12.50	
Max temp. (°C)	33.67	30.94	34.10	
Mean temp. (°C)	20.85	18.84	20.90	
Std dev. (°C)	1.05	1.24	1.32	

3.2 Algorithm Reimplementation

The Morris sensitivity analysis conducted in this thesis required executing the CWP detection algorithm across hundreds of parameter combinations, each applied to all three TIR orthophotos from Table 1-1, amounting to over 1000 individual algorithm runs in

total. Completing this within a feasible timespan required both a speed-optimized reimplementation and parallel execution on ZHAW’s High Performance Cluster (HPC). All benchmarking was performed directly on the cluster to ensure that optimisation gains carry over to the HPC infrastructure.

3.2.1 GDAL-Based Reimplementation

The reworked algorithm (v2) preserves the computational logic of the original implementation (v1) but differs fundamentally in how each step is mechanistically executed. Where v1 processes the raster slab-by-slab in memory using R loops and `terra/sf` functions, v2 replaces these with calls to the GDAL command line interface, operating directly on raster and vector files on disk rather than holding intermediate results in memory. Table 2-2 gives a high-level overview of v2; the tool or function used in each step is listed in the right-hand column.

Tab. 2-2: High-level overview of the reworked CWP detection algorithm, including the primary tool or function used for each step.

Step	Description	Tool / Function
1	Divide the river reach into n centerline segments; construct a wide zone polygon and a narrow reference-strip polygon for each segment.	<code>sf::st_buffer</code>
2	Rasterize the zone polygons to a raster matching the TIR raster resolution, using the segment id as burn-in value.	<code>gdal vector rasterize</code>
3	Round the TIR raster using a fixed rounding factor.	<code>gdal raster calc</code>
4	Extract the median temperature per segment from the rounded TIR raster within each reference strip; store in a look-up table keyed by segment id.	<code>exactextractr::exact_extract</code>
5	Reclassify the zone raster, replacing each segment id with its reference temperature from the look-up table, yielding the reference raster.	<code>gdal raster reclassify</code>
6	Subtract the rounded TIR raster from the reference raster; flag pixels where the difference exceeds ΔT , producing a binary CWP raster.	<code>gdal raster calc</code>
7	Polygonize the binary CWP raster into patch polygons.	<code>terra::as.polygons</code>
8	Filter patches by minimum area, compute per-patch temperature statistics, and apply a patch-level threshold check against the reference temperature.	<code>exactextractr::exact_extract</code>

The decision to heavily rely on GDAL was made as **terra** operations are sometimes not performant, and GDAL offers an often faster set of geocomputational tools, which can be called from R using `system()` or `sf::gdal_utils()` (Lovelace et al., 2019). Empirical testing in a prior coursework exercise also showed a significant speedup of GDAL-based raster operations compared to **terra** across a range of use cases (Dürig, 2026); this coursework was not peer-reviewed and is cited here only as a supporting empirical observation.

Operating directly on disk rather than in memory keeps the memory footprint small, which is important in an HPC setting where available memory can become a limiting factor under parallel execution. The drawback is that intermediate rasters can reach up to 25 GB in size; this was addressed by applying `COMPRESS=DEFLATE` with `PREDICTOR=2` when writing GeoTIFF files (GDAL/OGR contributors, 2026), which substantially reduces disk space consumption with negligible computational overhead.

`terra::extract()` was replaced with `exact_extract()` from the `exactextractr` package (steps 4 and 8 of Table 2-2), which extracts aggregated raster values considerably faster for large rasters (Mieno, 2022). The two functions differ in how partially covered cells are handled: `exact_extract()` weights cells by their coverage fraction (ISciences, LLC, 2026), whereas `terra::extract()` includes a cell only if its centre point falls within the polygon (Hijmans et al., 2024) – the behaviour used in v1. To approximate this, only cells with a coverage fraction of at least 50% were included in `exact_extract()`; this approximation is not exactly equivalent to **terra**’s centre-point rule.

3.2.2 Output Equivalence and Runtime Comparison

The resulting reimplementations were tested for equivalence in output by comparing the CWP polygons produced by v2 against those produced by v1 when deployed with the exact same set of parameters. Spatial agreement was assessed using the Jaccard similarity index:

$$J(A, B) = \frac{\text{area}(A \cap B)}{\text{area}(A \cup B)} \quad (2-1)$$

where A and B are the CWP polygon sets produced by v1 and v2 respectively. $J = 1$ indicates perfect agreement and $J = 0$ indicates no overlap. In addition, visual inspection in QGIS was carried out to identify any systematic spatial patterns in the disagreements. Table 2-3 shows the parameter combinations used in testing.

Runtime was compared across three step lengths (400, 800, and 1200 m), with all other parameters held constant (zone halfwidth = 60 m, buffer pixel count = 3 px, $\Delta T = 1.0^\circ\text{C}$, minimum patch area = 2 m^2 , rounding = 0.1°C , 8-connectivity enabled).

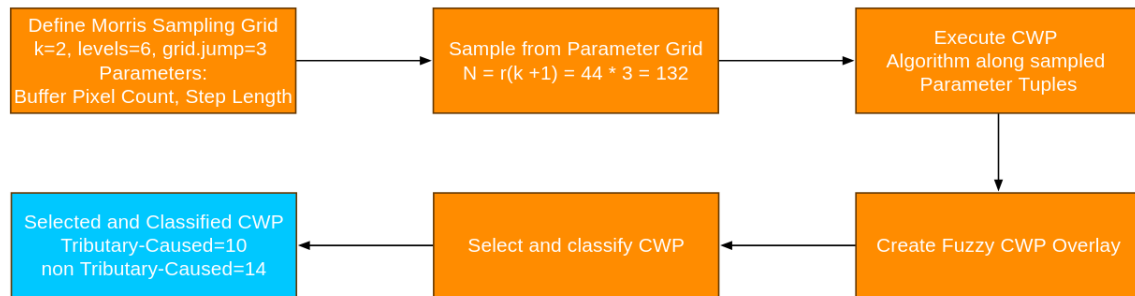
Tab. 2-3: Parameter combinations used for the output equivalence comparison between v1 and v2.

Parameter	Test 1	Test 2	Test 3
step_length	200	800	1200
zone_halfwidth_m	60	60	60
buffer_px	4	3	5
delta_T	0.5	1.0	1.5
min_patch_area_m2	2	3	2
round_to	0.1	0.1	0.1
connect_diagonals	TRUE	TRUE	TRUE

3.3 Sensitivity Analysis

With the algorithm optimized, sensitivity analysis was carried out to determine the sensitivity of the CWP detection algorithm with respect to the parameters buffer pixel count, step length, and temperature delta. As the step length sensitivity hypothesis concerns CWPs that are variable to the step length parameter specifically, a pre-selection step was first applied to identify CWPs that are actually susceptible to the step length parameter and to classify them by their likely cause, ensuring that the subsequent sensitivity analysis targets a relevant subset. Morris sensitivity indices were then computed for these CWPs with respect to their area. Finally, the results were visualized and a linear model relating step-length sensitivity to CWP area and temperature delta was fitted.

3.3.1 Determining and Classifying Step Length Sensitive CWP

**Fig. 3-2:** Visual overview of the steps involved in determining and classifying step-length-sensitive CWP. The result is a total of 24 classified CWP, where 10 are tributary-caused and 14 are non-tributary-caused.

First, a two-dimensional Morris sampling grid was set up spanning buffer pixel count and step length, with 6 levels per parameter (step length: 400–1200 m; buffer pixel count: 2–7 px), using the R package `sensitivity` to generate the sampling design. In this package, the step size is not set directly as a continuous value Δ , but rather through the `grid.jump` argument, an integer specifying the number of grid levels to be jumped when computing each elementary effect. A `grid.jump` of 3 was chosen because it corresponds to the recommended step size (see Section 2.2.3), which implements equal-probability sampling of the elementary effects, given by:

$$\Delta = \frac{p}{2(p-1)} = \frac{6}{2 \times 5} = \frac{3}{5} \quad (3-2)$$

Since the spacing between adjacent levels of the 6-level grid in the normalized unit space is $1/(p-1) = 1/5$, jumping 3 grid levels yields exactly $3 \times 1/5 = 3/5$, matching the recommended Δ . Figure 3-3 shows the grid with the jump distance drawn in. A total of 44 trajectories were then sampled, yielding $44 \times (2+1) = 132$ parameter tuples, each holding a unique combination of buffer pixel count and step length. The remaining parameters were held constant as listed in Table 3-4 to ensure that any variability in the model output is only attributed to either the step length or the buffer pixel count parameter.

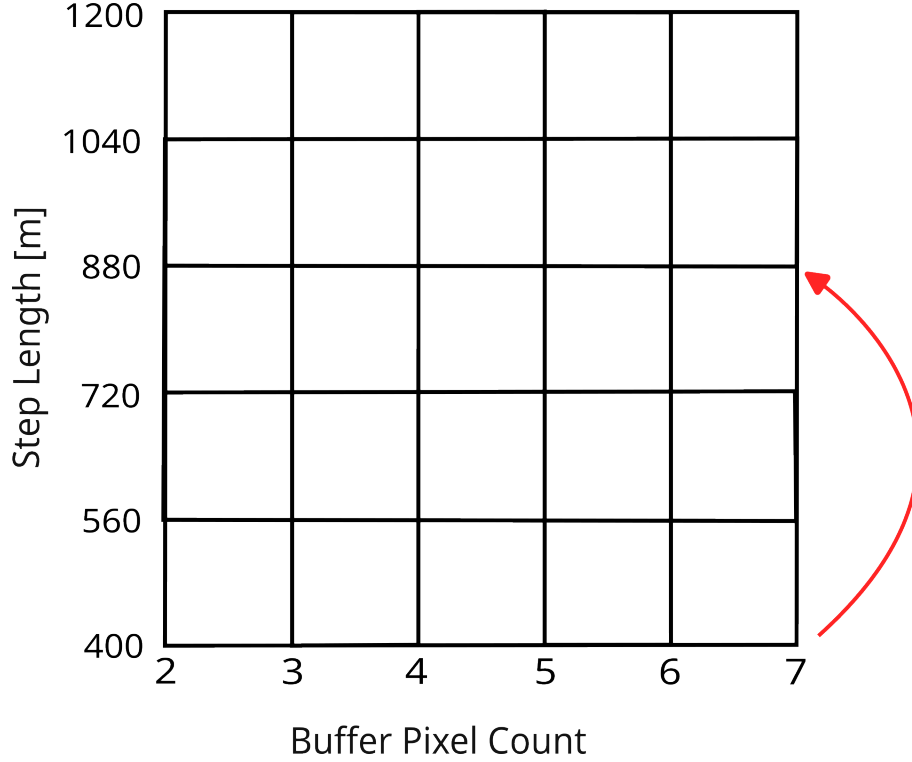


Fig. 3-3: Visual depiction of the Morris sampling grid, spanning step length and buffer pixel count. The arrow shows the jump distance chosen for the Morris experiment, corresponding to 3 grid levels.

Tab. 3-4: Parameter values held constant during the first Morris experiment, while step length and buffer pixel count were varied.

Parameter	Value
zone_halfwidth_m	60 m
delta_T	1.0 °C
min_patch_area_m2	2 m ²
connect_diagonals	TRUE
round_to	0.1

Next, the CWP algorithm was evaluated across all sampled parameter tuples in parallel using GNU Parallel (Tange, 2024). GNU Parallel orchestrates a pool of worker processes that operate concurrently. Each worker reads a parameter tuple from a shared

`parameters.txt` file and creates a uniquely named working directory based on the tuple identifier and parameter values – for example `job_1_7_880` for job 1 with `buffer_px = 7` and `step_length = 880` m. The worker then executes the CWP detection algorithm and writes all results to that directory before fetching the next available tuple. The cycle continues until all parameter tuples have been processed. This workflow is visualized in Figure 3-4.

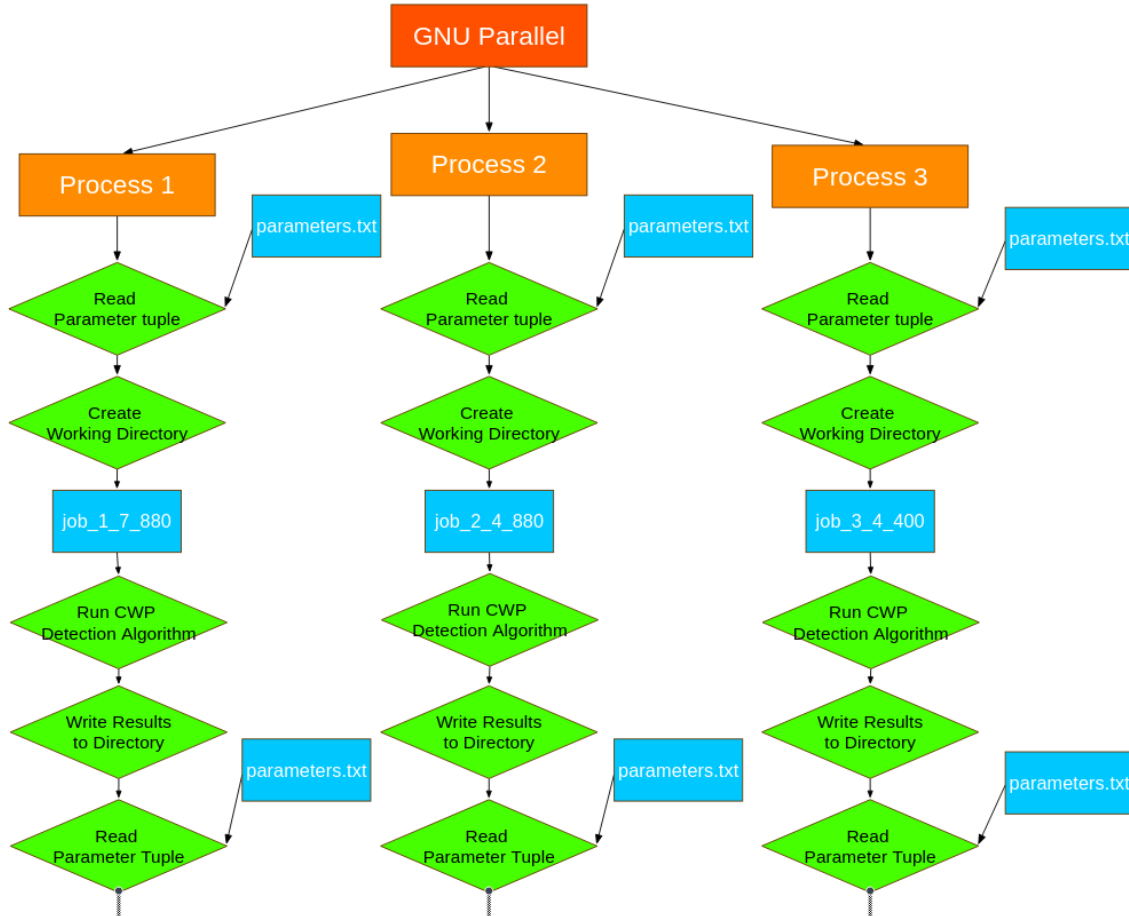


Fig. 3-4: An example setup showing the worker pool setup. Red: the GNU Parallel process. Orange: 3 worker processes in the pool. Green: actions performed by the worker processes. Blue: files and folders.

The CWP polygon layers from all runs were then rasterized at the TIR raster resolution, stacked into a multi-layer raster, and reduced to a single two-dimensional raster via pixel-wise mean, yielding a detection frequency raster. Each pixel in the detection frequency raster holds a score between 0 and 1, representing the fraction of runs in which that pixel was detected as part of a CWP. A pixel detected as CWP in every single run received a score of 1; a pixel never detected received a score of 0; pixels with intermediate scores were detected inconsistently across runs, reflecting sensitivity to the varied parameters. The construction of the detection frequency raster is illustrated in Figure 3-5.

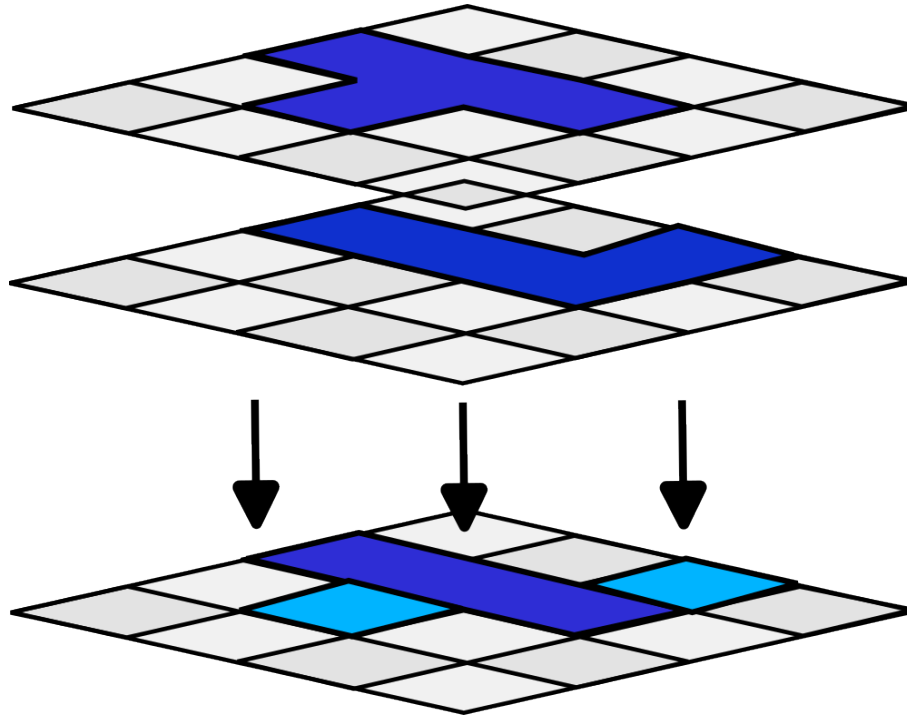


Fig. 3-5: Construction of the detection frequency raster. The upper two layers show individual CWP detection outputs from two runs using different parameter tuples. The lower raster shows the resulting detection frequency raster, with light blue pixels indicating lower detection frequency than dark blue pixels.

Finally, from the detection frequency raster, patches that were clearly variable and clearly attributable to either a tributary inflow or its absence were selected and classified as tributary-caused or non-tributary-caused based on visual interpretation. Variability was assessed by inspecting the contrast of the detection frequency raster: patches showing a wide range of scores – from black to white – across their spatial extent were considered variable. Tributary attribution was then determined by proximity to a tributary inflow; patches lying adjacent to a visible tributary confluence were classified as tributary-caused, while patches with no nearby tributary were classified as non-tributary-caused. Figure 3-6 shows examples of the four possible classification outcomes.



(a) Variable, tributary-caused. Coloring indicates variability, and the patch lies adjacent to a tributary inflow.



(b) Variable, non-tributary-caused. Coloring indicates variability, but no tributary inflow is present nearby.



(c) Non-variable, tributary-caused. The patch shows little variability in the detection frequency raster despite proximity to a tributary inflow.



(d) Non-variable, non-tributary-caused. The patch shows little variability and no tributary inflow is present nearby.

Fig. 3-6: Examples of the four classification outcomes from the decision tree in Figure ??: variable vs. non-variable patches (rows), further split by tributary attribution (columns).

Tributary inflows were identified using OpenStreetMap and SwissImage in combination. OpenStreetMap allows tributaries obscured by vegetation to be identified, while SwissImage covers tributaries that were not mapped in OpenStreetMap – neither source can be assumed complete on its own. Multiple closely adjacent CWPs were grouped and treated as a single patch, as CWPs in close spatial proximity share the same local generating conditions and will therefore exhibit highly similar sensitivity – treating them as separate observations would violate the assumption of independence and skew the sensitivity indices towards conditions prevailing at that particular spot. For each of the now classified CWP, the bounding box was obtained indicating their location.

3.3.2 Computing Morris Sensitivity Indices with respect to CWP Area

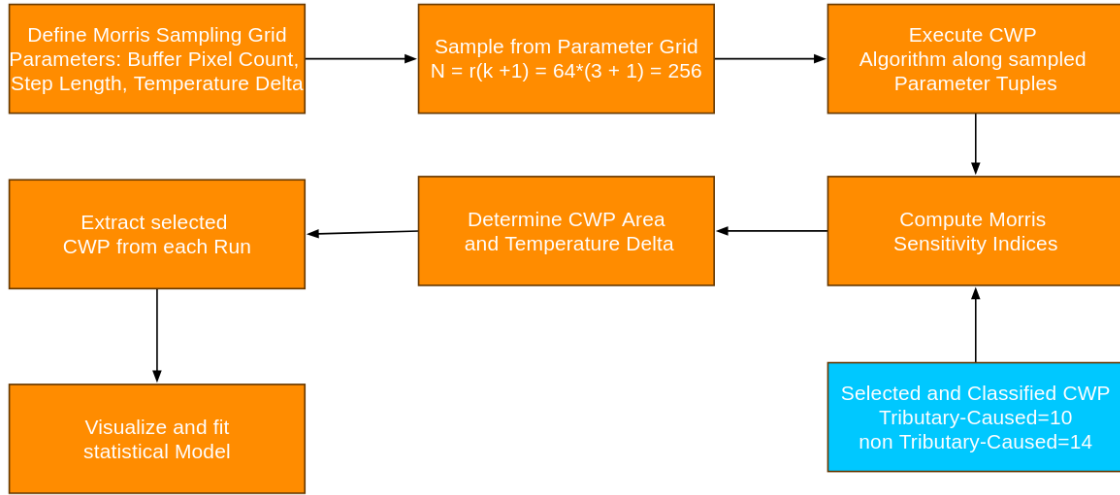


Fig. 3-7: Visual overview of the steps involved in computing Morris sensitivity indices with respect to CWP area. The selected and classified CWP locations flow into the process at step 4.

With the 24 CWP classified, the second stage quantified their sensitivity to the three algorithm parameters: buffer pixel count, step length, and temperature delta. Where the first experiment varied only two parameters to produce the detection frequency raster, the second experiment varied all three and used the per-location CWP area as the scalar output fed into the Morris sensitivity computation.

First, a three-dimensional Morris sampling grid spanning buffer pixel count, step length, and temperature delta was set up, as depicted in Figure 3-8. A `grid.jump` of 3 was again chosen, implementing equal-probability sampling of the elementary effects as described in Section 2.2.3. A total of 64 trajectories were then sampled from the grid, yielding $64 \times (3+1) = 256$ parameter tuples (Equation 2-3). The algorithm was executed in parallel using the same orchestration as in the first experiment, with the remaining parameters held constant at the same values as listed in Table 3-4.

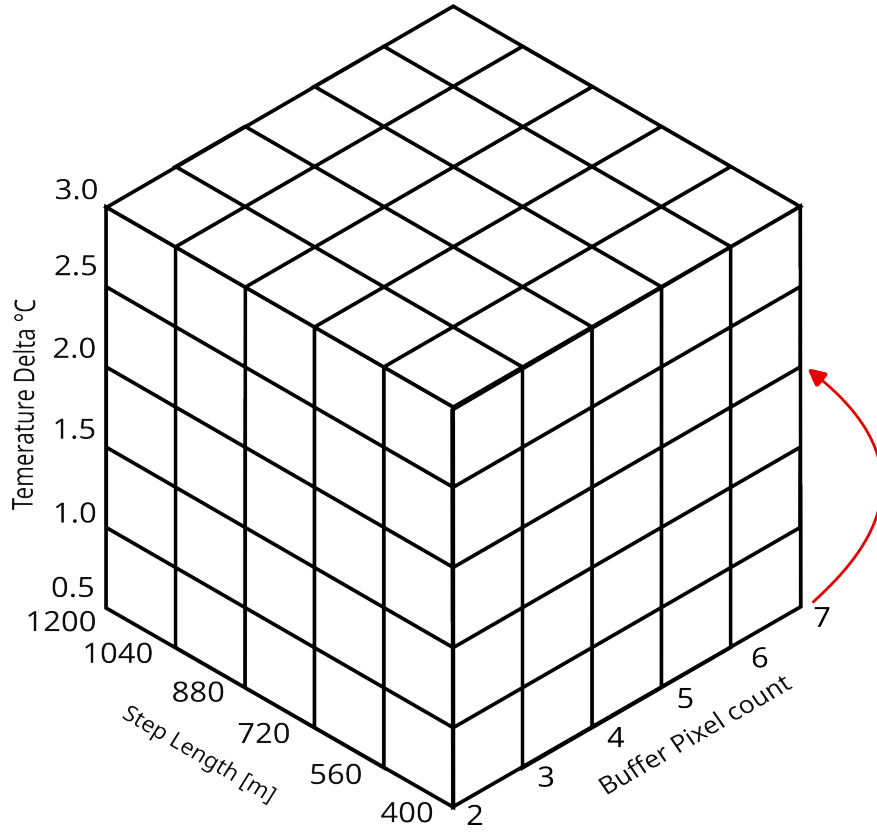


Fig. 3-8: Visual depiction of the second Morris sampling grid, spanning step length, buffer pixel count, and the temperature delta threshold. The red arrow indicates the jump distance chosen.

Next, for each of the 24 CWP bounding boxes and each completed Morris run, the CWP polygons lying inside the bounding box were selected. The total CWP area and the area-weighted mean temperature delta were then computed from these polygons. If a given parameter tuple produced no CWP detection within the bounding box, both statistics were set to zero. The computation and the resulting table structure are illustrated in Figure 3-9.

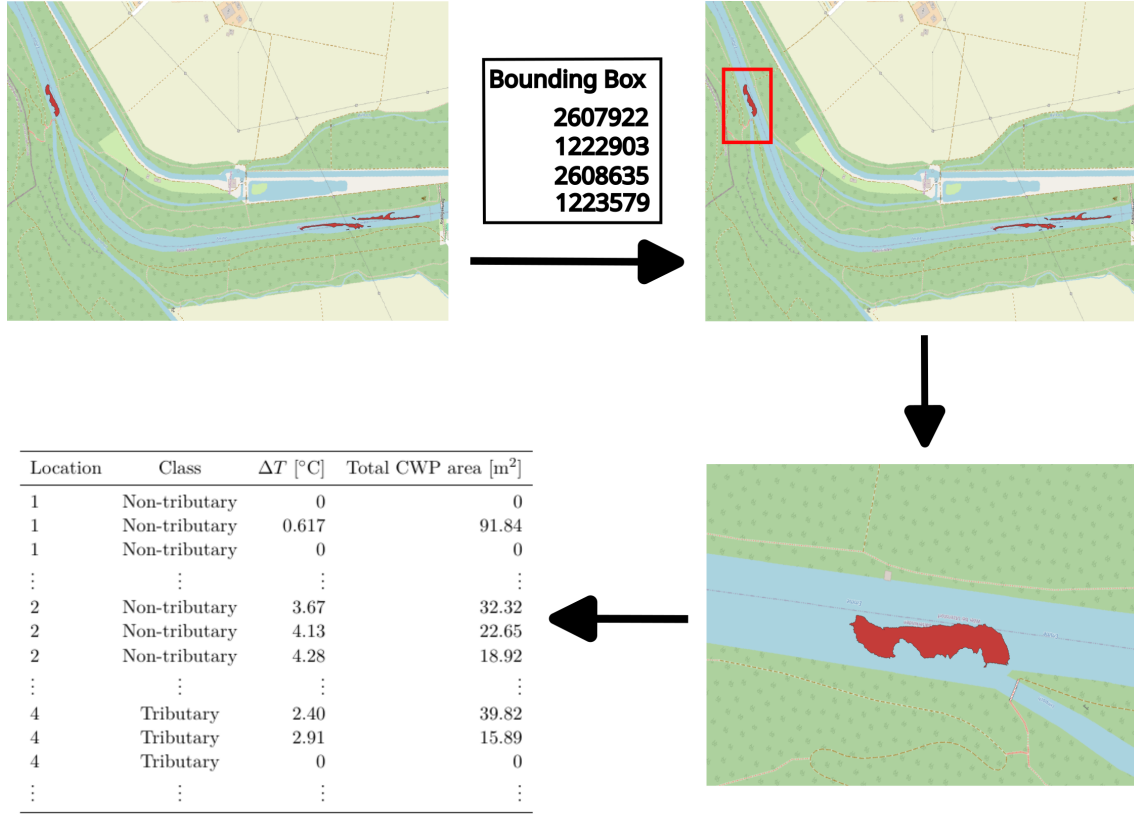


Fig. 3-9: Visual representation of the bounding box computation. A CWP located in one of the 24 bounding boxes is selected, then aggregation is performed. The resulting statistics can be captured in a table based format together with aggregation results from other bounding boxes.

Finally, the total CWP area values were normalised per bounding box using min-max scaling into $[0, 1]$. This ensured comparability across bounding boxes: without normalisation, bounding boxes with inherently larger CWP would produce larger absolute area differences, inflating their sensitivity indices relative to smaller patches. After scaling, a value of 1 corresponded to the maximum area detected for that bounding box across all runs and a value of 0 to an empty bounding box. The Morris indices μ^* , and σ were then computed with respect to the normalised area values using the R package `sensitivity`.

Two sets of boxplot visualisations were produced. The first displayed μ^* and σ with respect to step length, stratified by CWP class, to allow a direct comparison of sensitivity between tributary-caused and non-tributary-caused CWPs. The second displayed μ^* for all three varied parameters pooled across both classes, providing an overview of their relative influence on CWP area.

Additionally, a linear model was fitted to assess whether CWP area and temperature delta are associated with step-length sensitivity:

$$\mu_{\text{step length}}^* = \beta_0 + \beta_1 \cdot \overline{\text{area}} + \beta_2 \cdot \overline{\Delta T} + \varepsilon \quad (3-3)$$

where $\mu_{\text{step length}}^*$ is the Morris sensitivity index for step length, $\overline{\text{area}}$ and $\overline{\Delta T}$ are the mean total CWP area and mean temperature delta across all Morris runs for each CWP location, and ε is the residual error term, capturing the portion of variability in $\mu_{\text{step length}}^*$

that cannot be explained by mean area and mean temperature delta alone. The model was fitted using R's `lm()` across all 24 locations, pooling both classes.

3.4 Algorithm Modification

Two findings from [Tonolla and Antonetti \(2026\)](#) informed specific algorithmic decisions in the design of v3.

First, the buffer pixel count was found to have little influence on the detected CWP area, qualifying it as a candidate for factor fixing – that is, it can be set to a fixed constant and removed as a free parameter without meaningfully affecting the algorithm's output. In v3, the buffer pixel count is therefore internalized and set to a constant value of 3, which lies within the range of values examined by [Tonolla and Antonetti \(2026\)](#).

Second, the step length parameter was identified as having a non-negligible influence on the detected CWP area. A plausible mechanistic explanation for this sensitivity is that CWPs spanning a slab boundary are compared against two different reference temperatures – one per slab – which can produce artificial truncations in the detected patch extent. This motivated a fundamental redesign of the reference temperature computation in v3, replacing the slab-based approach with a spatially continuous IDW-based scheme that does not depend on a step length parameter at all. The temperature delta parameter is retained as a free parameter, as different thresholds are used in the literature to define CWPs (see [2.4](#)) and the user must therefore retain the ability to set it according to their chosen definition.

3.4.1 Operationalisation of Surrounding Stream Temperature

In v1 and v2, the reference temperature for each pixel is determined by the median temperature of the slab segment to which that pixel belongs. This approach has two consequences: it requires a user-defined step length, and it produces a reference temperature field that is piecewise constant – uniform within each slab and discontinuous at slab boundaries. CWPs that happen to span a slab boundary are therefore compared against two different reference temperatures, which can produce artificial cutoffs in the detected patch extent.

In v3, this is replaced by an IDW scheme. Rather than dividing the reach into fixed slabs, reference temperature sample points are obtained by sampling multiple step length values from a uniform distribution and extracting the median temperature at each resulting segment's visual center. These sample points, pooled across all sampled step lengths, are then used to compute a spatially continuous reference temperature raster via IDW, where each cell's reference temperature is a distance-weighted average of the surrounding sample point temperatures. This guarantees that the reference temperature is a spatially continuous field, and that no sharp cutoffs at slab boundaries can occur.

3.4.2 Implementation Detail

The implementation of v3 follows the structure of v2 closely, differing only in the computation of the reference temperature raster (steps 1–5 below, replacing steps 1–5 of [Table 2-2](#)). The full reference temperature computation proceeds as follows:

Tab. 4-5: High-level overview of the v3 CWP detection algorithm. Steps 1–5 replace the corresponding steps of Table 2-2; steps 6–8 are identical to v2.

Step	Description
1	Sample 20 candidate step length values from a uniform distribution with bounds 500–3000 metres.
2	For each candidate step length, divide the river centerline into n segments; construct a narrow 3-pixel-wide reference strip polygon for each segment; extract the median temperature of each segment from the TIR raster; compute the visual centre of each strip polygon and assign the corresponding median temperature to this point.
3	Pool the reference points (location + temperature) from all 20 step length values into a single combined set of IDW sample points.
4	Initialise a new raster with the same extent and resolution as the TIR raster. For each cell: if the corresponding TIR cell is NaN, set to NaN; otherwise compute a distance-weighted reference temperature via IDW over the pooled sample points, using a decay power of 2.
5	Round the TIR raster using a fixed rounding factor.
6	Subtract the rounded TIR raster from the reference raster; flag pixels where the difference exceeds ΔT , producing a binary CWP raster.
7	Polygonize the binary CWP raster into patch polygons.
8	Filter patches by minimum area; compute per-patch temperature statistics; apply a patch-level threshold check against the reference temperature.

Step 4 is implemented in C++ and imported into R via the **Rcpp** package, as the per-pixel IDW computation was identified as the primary computational bottleneck.

3.4.3 Reduction of Parameters

Table 4-6 summarises the status of each free parameter in v3 relative to v2. The zone halfwidth elimination implies a constraint: v3 requires a TIR raster with an exact water mask, with non-river pixels set to NaN.

Tab. 4-6: Parameter status in v3 relative to v2.

Parameter	Status in v3	Reason
buffer_px	Fixed at 3	Non-influential; factor fixing applied
step_length	Removed	Replaced by IDW sampling over a range of step lengths
zone_halfwidth_m	Removed	Reference raster matched to valid TIR extent
delta_T	Retained	User-defined threshold; varies across CWP definitions
min_patch_area_m2	Retained	User-defined filter
round_to	Retained	User-defined rounding factor
connect_diagonals	Retained	User-defined connectivity rule

3.5 Usage of AI

The large language models Claude Sonnet 4.6 and Claude Opus 4.6 (Anthropic) were used as aids in this project. Detailed insight into the usage is given in Appendix A.

3.6 Algorithm Reimplementation

The following two subsections present the runtime benchmarking and output equivalence results for v1 and v2.

3.6.1 Runtime Comparison

Figure 6-10 shows the execution time of v1 and v2 as a function of step length. The execution time of v2 is approximately 0.5 hours across all three tested step lengths, remaining largely constant. In contrast, v1 shows execution times ranging from approximately 2 hours at a step length of 400 m to approximately 1.25 hours at 1200 m, with execution time increasing as step length decreases (i.e., as the number of segments increases).

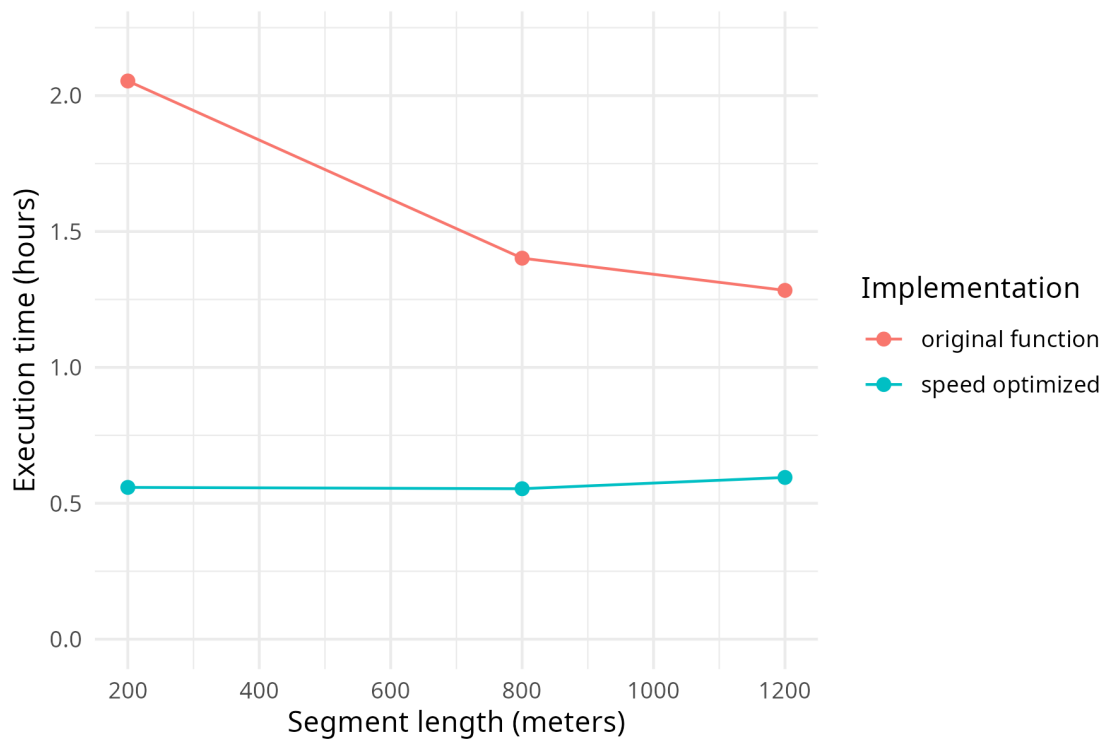
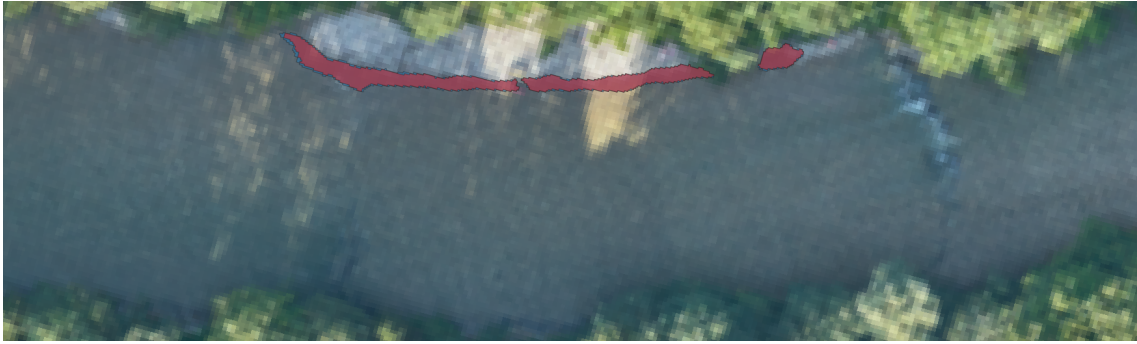


Fig. 6-10: Execution time of v1 and v2 as a function of step length. In red the original CWP detection algorithm (v1) in blue the speed optimized algorithm (v2).

3.6.2 Output Equivalence

The three comparison runs achieved Jaccard similarities of 0.936, 0.956, and 0.976, indicating generally good spatial agreement between v1 and v2. Figure 6-11 illustrates the range of agreement observed. Panel (a) shows a representative example of general agreement, with discrepancies confined to a small number of boundary pixels. Panel (b) shows a more pronounced boundary mismatch. Panel (c) shows an example of complete spatial agreement, where the two polygon sets are visually indistinguishable.



(a) General agreement. Discrepancies are confined to patch boundaries.



(b) More pronounced disagreement. Boundary mismatches between v1 and v2 are more evident.



(c) Complete agreement. v1 and v2 polygons are spatially indistinguishable.

Fig. 6-11: Comparison of CWP polygon outputs between v1 (blue) and v2 (red) across the three validation runs. Polygons are overlaid at 50% transparency; areas of agreement appear as a blend of both colours, while areas covered by only one algorithm appear in that algorithm's colour alone.

3.7 Sensitivity Analysis

The Morris sensitivity analysis results are presented below across three complementary views, progressing from the specific hypothesis regarding step-length sensitivity to the broader parameter ranking and the relationship between CWP area and sensitivity.

3.7.1 Sensitivity of CWP Area to Step Length

Figure 7-12 shows the Morris sensitivity indices μ^* and σ with respect to the step length parameter, stratified by CWP class. All non-tributary-caused CWP show μ^* values below 0.2, indicating low sensitivity to the step length parameter across this class. The tributary-caused CWP show a wider spread in μ^* : some tributary-caused CWP have sensitivity values comparable to the non-tributary-caused group, while others show substantially higher values of up to 0.3–0.6. The median μ^* values of the two classes are similar, with the primary difference lying in the upper tail of the tributary-caused distribution. The σ values show a comparable pattern, with greater spread in the tributary-caused group.

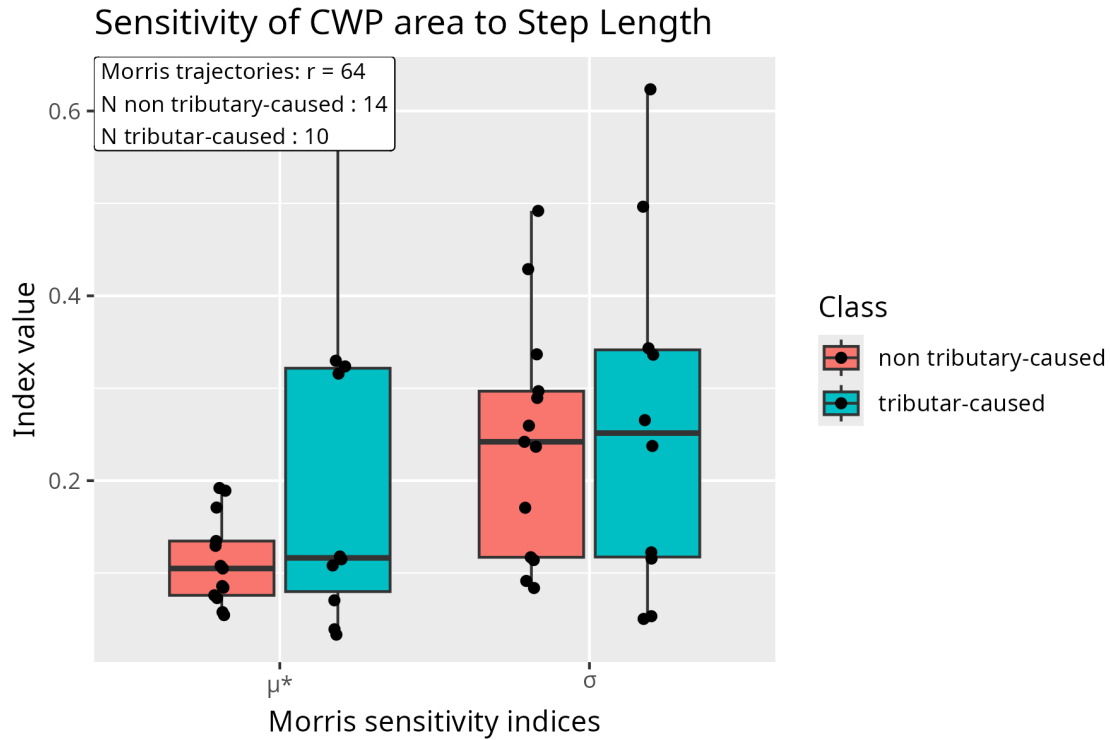


Fig. 7-12: Morris sensitivity indices (μ^* and σ) for the `step_length` parameter with respect to cold water patch area, stratified by patch class (red: tributary-caused, $n = 10$; blue: non-tributary-caused, $n = 14$). Boxplots show the distribution across bounding boxes.

3.7.2 Overall Sensitivity across all Parameters

Figure 7-13 displays the μ^* and σ values for all 24 CWP locations with respect to the three parameters varied in the second Morris experiment. The buffer pixel count shows consistently low μ^* values across all locations, with the entire distribution concentrated near zero, and a similarly negligible σ , indicating neither a meaningful main effect nor relevant interactions with the other parameters. In contrast, both step length and temperature delta show distributions that deviate clearly from zero for both μ^* and σ . The temperature delta parameter shows the highest μ^* values overall, suggesting it has the strongest influence on the detected CWP area. For both step length and temperature delta, σ is of comparable magnitude to μ^* , pointing to the presence of nonlinearity or interactions with the other parameters for both.

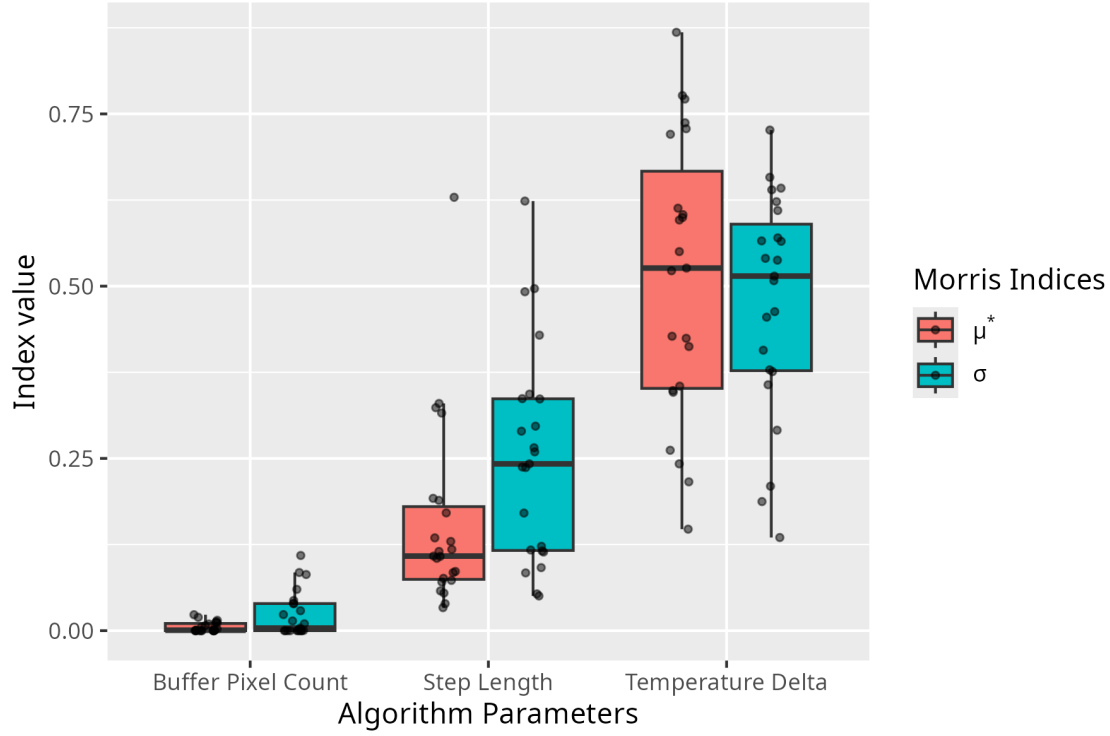


Fig. 7-13: μ^* and σ sensitivity indices for buffer pixel count, step length, and temperature delta across all classified cold water patches ($N = 24$ locations, $r = 64$ trajectories).

3.7.3 Relationship between CWP Area and Step Length Sensitivity

Figure 7-14 shows the 24 CWP locations along three dimensions simultaneously: mean CWP area averaged across all Morris runs (x-axis), μ^* with respect to the step length parameter (y-axis), mean temperature delta averaged across all Morris runs (color), and CWP class (shape). Locations with larger mean CWP area tend to show higher μ^* values. No clear pattern between mean temperature delta and μ^* is visible from the colouring.

Table 7-7 shows the results of the linear model. The mean area predictor is highly significant ($p < 0.001$) with a positive slope, consistent with the visual pattern in Figure 7-14. The mean temperature delta predictor is not significant ($p = 0.556$). The model explains a substantial proportion of variance ($R^2 = 0.611$, Adjusted $R^2 = 0.572$).

Tab. 7-7: Regression results for μ^* (step length) predicted by mean CWP area and mean ΔT , fitted across all 24 annotated locations.

Predictor	Estimate	SE	t	p
(Intercept)	9.024×10^{-2}	3.094×10^{-2}	2.92	0.009
Mean area	1.578×10^{-4}	2.839×10^{-5}	5.56	< 0.001
Mean ΔT	-1.041×10^{-2}	1.739×10^{-2}	-0.60	0.556

Note. $R^2 = 0.611$, Adjusted $R^2 = 0.572$, $F(2, 20) = 15.73$, $p < 0.001$. Residual standard error = 0.088 on 20 degrees of freedom (one observation deleted due to missingness).

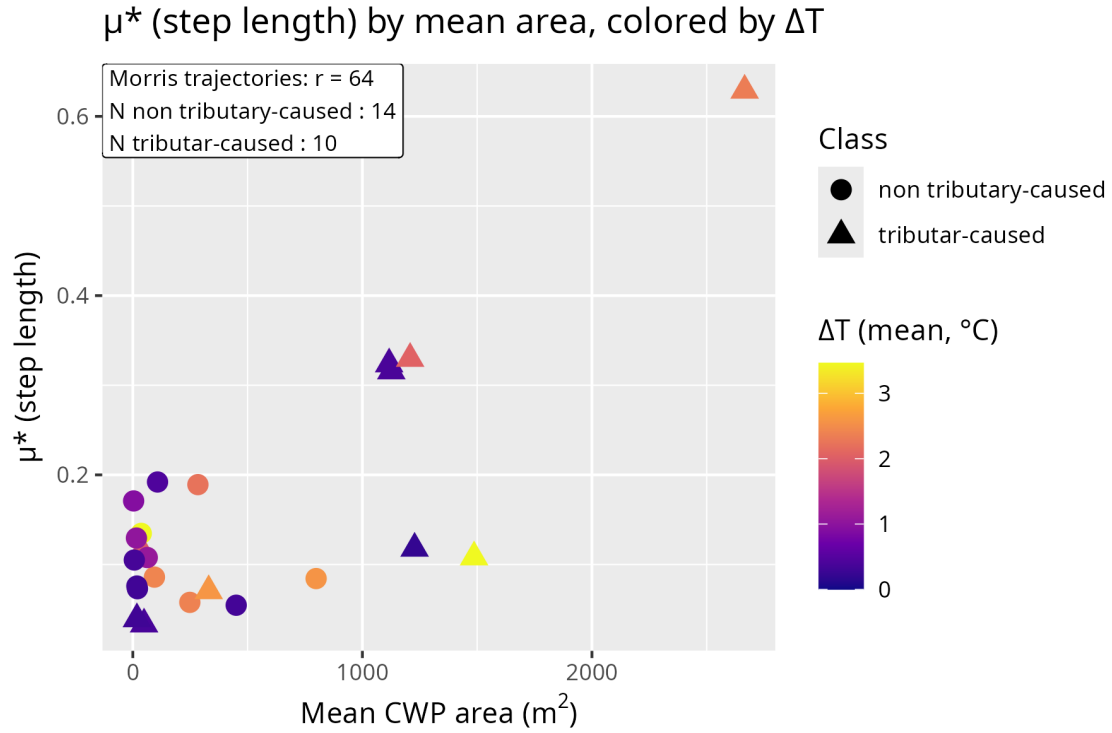


Fig. 7-14: μ^* (step length) versus mean CWP area for each of the 24 annotated locations. Color indicates the mean temperature delta to the surrounding river ($\overline{\Delta T}$, averaged across all Morris runs); shape indicates CWP class (circle: non-tributary-caused; triangle: tributary-caused).

3.8 Algorithm Modification

The CWP polygons produced by v3 were compared against those of v1 using the same Jaccard similarity metric applied in Section 3.2.2. The Jaccard similarities obtained were 0.325, 0.334, and 0.287 – substantially lower than the values of 0.936, 0.956, and 0.976 obtained for v2 versus v1. Figure 8-15 illustrates three qualitative differences observed in the v3 detections relative to v1.



(a) A v1 polygon ends abruptly mid-reach; the corresponding v3 polygon extends beyond that point.



(b) Two separate v1 polygons are detected as a single contiguous patch in v3.



(c) A CWP detected by v3 that is entirely absent in v1.

Fig. 8-15: Qualitative comparison of CWP detections between v1 and v3 at three example locations. v1 polygons and v3 polygons are overlaid at 50% transparency; areas covered by both algorithms appear as a blend of the two colours, while areas covered by only one algorithm appear in that algorithm's colour alone.

Panel (a) shows a case where the v1 polygon ends abruptly mid-reach while the v3 polygon extends beyond that point and tapers off gradually. Panel (b) shows a location where two separate v1 polygons are merged into a single contiguous patch in v3. Panel (c) shows a CWP detected by v3 that is entirely absent in v1.

Chapter 4

Discussion

4.1 Algorithm Reimplementation

This section discusses the results of the output equivalence comparison and the runtime benchmarking between v1 and v2, and assesses whether the sensitivity analysis findings derived from v2 can be transferred to v1.

4.1.1 Difference in CWP Polygons between v1 and v2

The CWP polygons produced by v1 and v2 are in general agreement with respect to location and spatial extent, as reflected by Jaccard similarities of 0.936, 0.956, and 0.976 across the three validation runs. The observed discrepancies are confined to the pixel level – typically affecting a small number of pixels at patch boundaries – and no systematic spatial bias is apparent.

The most likely source of these discrepancies is the difference in how `terra::extract()` and `exactextractr::exact_extract()` handle partially covered pixels: the 50% coverage-fraction threshold used to approximate `terra`’s centre-point inclusion rule is not exactly equivalent, and small differences in extracted reference temperatures propagate to pixel-level differences in the binary CWP raster.

The high Jaccard similarities, combined with the observation that the discrepancies can be attributed to a minor extraction artefact rather than a fundamental algorithmic difference, support the conclusion that v2 replicates the behaviour of v1 to a degree sufficient to transfer sensitivity findings.

4.1.2 Speed Comparison

The optimized algorithm (v2) shows substantially shorter execution times than v1 across all three tested segment lengths, with the performance advantage most pronounced at shorter step lengths. While v2 maintains an approximately constant execution time of around 0.5 hours regardless of step length, v1’s execution time increases as step length decreases, ranging from approximately 1.25 hours at 1200 m to approximately 2 hours at 400 m.

This difference in scaling behaviour is consistent with a mechanistic explanation rooted

in how the two implementations process the river reach. v2 is loop-free: GDAL operates on the full raster in a single pass, and each pixel undergoes a fixed number of operations regardless of how many segments the river reach is divided into. v1, by contrast, processes the reach one slab at a time using R loops. Each loop iteration incurs a fixed overhead independent of slab size, plus a variable computation proportional to the raster area of that slab. At shorter step lengths the reach is divided into more, smaller slabs, shifting the ratio of overhead-to-useful-work unfavourably and increasing total execution time. It should be noted that this mechanistic account is a hypothesis and was not formally tested within this work.

v2 is faster than v1 across the entire parameter range tested, and the performance advantage is largest at the shorter step lengths (400 m) where v1 is slowest – which fall within the Morris sampling grid used in this work.

4.2 Sensitivity Analysis

This section interprets the Morris sensitivity analysis results, first addressing the specific hypothesis regarding step-length sensitivity in tributary-caused versus non-tributary-caused CWP, and then discussing the overall parameter sensitivity ranking across all three varied parameters.

4.2.1 Step Length Sensitivity by CWP Class

The boxplot comparison of μ^* values between tributary-caused and non-tributary-caused CWP shows a tendency consistent with H1: tributary-caused CWP display a wider spread in step-length sensitivity, with a higher third quartile and higher maximum values than the non-tributary-caused group. However, a closer inspection reveals that this pattern is not uniform within the tributary-caused class – some tributary-caused CWP show sensitivity values comparable to the non-tributary-caused group, while others are substantially more sensitive. This within-class heterogeneity suggests that tributary presence alone is not a sufficient condition for high sensitivity to the step length parameter, and that H1 as stated does not fully capture the situation. At the same time, the data do not support H0 either: the non-tributary-caused CWP consistently show low μ^* values, and the elevated sensitivity observed in a subset of tributary-caused CWP is unlikely to be entirely coincidental. The true picture lies between the two hypotheses.

The scatter plot and linear model together point to CWP area as the more proximate driver of step-length sensitivity. Across both classes, locations with larger mean CWP area tend to show higher μ^* values, and this relationship is approximately linear. The linear model confirms this: the mean area predictor is highly significant ($p < 0.001$) with a positive slope, while the mean temperature delta predictor is not significant ($p = 0.556$).

A mechanistic explanation for the area-sensitivity relationship follows naturally from the structure of the algorithm. CWPs in a river system tend to be elongated rather than wide, since the river channel constrains lateral extent while longitudinal extent is less bounded. Large CWP are therefore predominantly long CWP. A long CWP has a higher probability of spanning the boundary between two adjacent slabs, and when this occurs, the two parts of the patch are compared against different reference temperatures – one computed from each slab’s reference strip. If those reference temperatures differ (which is

more likely near a tributary inflow, where sharp longitudinal temperature gradients exist), the flagging threshold is effectively applied inconsistently across the patch, introducing variability in the detected CWP extent that is directly attributable to the step length parameter. Shorter or smaller CWP are more likely to fall entirely within a single slab, and are therefore less exposed to this effect.

Tributary inflows are then best understood not as a direct cause of step-length sensitivity, but as one mechanism that can produce large, elongated CWP – by introducing substantial volumes of cold water that generate extended thermal anomalies along the river. It also means that the data cannot speak to whether a large non-tributary-caused CWP would show comparable sensitivity to an equally large tributary-caused CWP, since no non-tributary-caused CWP larger than approximately 1000 m² is present in the dataset.

Interestingly, CWP area and mean temperature delta appear to be largely unrelated in the data, despite the intuitive expectation that a colder tributary might produce a larger thermal anomaly. This could suggest that tributary discharge plays a more important role than temperature difference in determining CWP size – a large, moderately cold tributary may produce a larger CWP than a small but very cold one. Discharge data were not collected during the survey, but could be a valuable addition in future work.

Neither H0 nor H1 is fully supported by the data. H0 is too strong a rejection – tributary presence is associated with a tendency towards higher step-length sensitivity. However, H1 overstates the role of tributary presence as a direct determinant of sensitivity. The data suggest that the more proximate driver is CWP size: large CWP, which are disproportionately but not exclusively tributary-caused, are more susceptible to the step length parameter because they are more likely to span slab boundaries and therefore be exposed to inconsistent reference temperatures.

4.2.2 Overall Sensitivity across all Parameters

The overall μ^* distributions across all 24 annotated locations reveal a clear ranking of parameter influence with respect to detected CWP area. The buffer pixel count is consistently non-influential, with μ^* values close to zero across all locations. The corresponding σ values are similarly low, suggesting that the elementary effects of buffer pixel count are both small in magnitude and consistent across the parameter space – indicating neither a strong main effect nor meaningful interactions with the other parameters. This finding is consistent with [Tonolla and Antonetti \(2026\)](#), who report the same result in their sensitivity analysis of the CWP detection algorithm. Given this replication, the strategy of factor fixing can be applied: the buffer pixel count can be set to a fixed constant value, effectively removing it as a free parameter of the algorithm.

The temperature delta parameter shows the highest μ^* values overall, indicating that it is the most influential of the three parameters with respect to CWP area. This is also consistent with the findings of [Tonolla and Antonetti \(2026\)](#), who likewise identify temperature delta as the dominant parameter. The elevated σ values for temperature delta suggest that its elementary effects vary across the parameter space, pointing to the presence of nonlinearity or interactions with the other parameters. This is ecologically plausible: the relationship between the flagging threshold and the detected CWP area is unlikely to be linear, as small changes in temperature delta near the boundary of what constitutes a detectable thermal anomaly could produce disproportionately large changes in detected area, while changes far from this boundary may have little effect.

The step length parameter occupies an intermediate position, with μ^* values clearly above zero but consistently below those of temperature delta. Like temperature delta, step length also shows elevated σ values, suggesting interactions with the other parameters or nonlinear behaviour.

4.3 Algorithm Modification

The replication of the buffer pixel and temperature delta findings from [Tonolla and Antonetti \(2026\)](#) within the present work, using an independently implemented Morris experiment on a different dataset, lends additional credibility to both sets of results and suggests that these sensitivity patterns are robust properties of the CWP detection algorithm rather than artefacts of a particular study context. This agreement between the two analyses provides a stronger basis for the design decisions underlying v3 than either analysis alone would afford.

Based on these findings, three modifications were implemented in v3, as described in Section 3.4: the buffer pixel count was fixed by factor fixing, the slab-based reference temperature computation was replaced by an IDW scheme removing the step length dependency, and the temperature delta parameter was retained as a user-defined input. The evaluation presented in Section 3.8 shows that v3 produces outputs that differ substantially from v1 in terms of Jaccard similarity, which is expected given the fundamental redesign of the reference temperature step. Qualitatively, however, v3 eliminates the slab-boundary cut-off artefacts identified as a direct consequence of step-length sensitivity, producing more contiguous and naturally extending CWP detections. Research question 3 can therefore be answered affirmatively: the identified parameter sensitivities can be addressed through targeted algorithmic modifications without sacrificing output plausibility.

4.4 Future Improvements

The following subsections identify three methodological limitations of the present work and outline directions in which future studies could address them.

4.4.1 Subjective Selection and Classification of CWP

Both the selection and classification of the CWP were carried out subjectively based on visual inspection of the fuzzy overlay raster. This approach is prone to confirmation bias on two levels. First, the judgment about variability and the judgment about class membership were both made with the study hypothesis already in mind. CWPs that were selected are therefore likely those that were most easily attributable to one of the two classes – patches that clearly appeared variable and clearly appeared tributary-caused or non-tributary-caused. More nuanced cases, where variability was ambiguous or class attribution uncertain, were left out. The resulting dataset is consequently biased towards extreme and obvious cases, and may not be representative of the broader population of CWPs in the Emme system. Second, given that H1 of research question 2 was considered plausible prior to the analysis based on prior observations, the selection process may have been further skewed towards patches that would support H1, even if this influence was not conscious.

Two methodological improvements could reduce these limitations in future work. First, the variability assessment and CWP selection could be replaced by a fully algorithmic approach, removing human judgment from the process. Caution is however warranted: upon defining the rules of a process-based algorithm, the analyst may inadvertently encode the same biases that affect manual selection. Similarly, a data-driven approach would need to learn from a ground truth dataset that is itself produced by human labeling, meaning the bias present in the labeling process would be inherited by the model. Second, if manual classification cannot be avoided, inter-rater reliability assessment should be included: multiple independent observers, each blind to the study hypothesis, could classify the same set of candidate CWPs, and the level of agreement between raters – quantified for example using Cohen’s κ – would provide an empirical measure of classification reliability. Such a dataset could further serve as ground truth for training a data-driven classification algorithm, combining the transparency of manual labeling with the scalability of an automated approach.

4.4.2 Sample Size

With only $n = 24$ annotated CWP locations, the dataset is small by the standards of statistical inference. While the significant area–sensitivity relationship detected by the linear model suggests that the effect is strong enough to be visible at this sample size, the risk of chance findings cannot be excluded. A larger sample would increase statistical power and reduce the probability that the observed relationships are artefacts of the specific locations selected.

4.4.3 Hidden Parameter in v3

Upon closer inspection, v3 introduces a parameter that was not explicitly treated as a free parameter during the development of the algorithm: the decay power of the inverse distance weighting scheme, which governs how quickly the influence of a reference point diminishes with distance. In the current implementation this value is fixed at 2, which is a common default, but the choice is arbitrary and its influence on the detected CWP area has not been assessed. Strictly speaking, this means that the parameter reduction achieved by v3 is incomplete: the decay power has effectively been fixed by convention rather than by a principled sensitivity analysis. A dedicated sensitivity analysis of the IDW decay power would be needed to either justify fixing it at its current value through factor fixing, or to expose it as a free parameter that the user should be able to configure.

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Chapter 5

Appendix

- Appendix A: Catalog of classified CWP
- Appendix B: Statement of Reproducibility and Collaboration
- Appendix C: List of Generative AI Systems

Appendix C: List of Generative AI Systems

The following generative AI systems and AI-assisted tools were utilized in the context of this work:

System Tool	/	Provider & Version	Purpose of Use
Claude Sonnet 4.6		Anthropic (2026) Version Sonnet 4.6	• Stylistic improvement of text • Spelling correction • Code debugging and coding assistance (no agentic coding) • L^AT_EX assistant (debugging and formatting) • Finding relevant literature
Claude Opus 4.6		Anthropic (2026) Version Opus 4.6	• Complex debugging tasks • Complex writing assistance tasks such as assuring cross-chapter naming consistency

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